Machine Learning

Chapitre 1 : Introduction

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Types of Machine Learning Systems

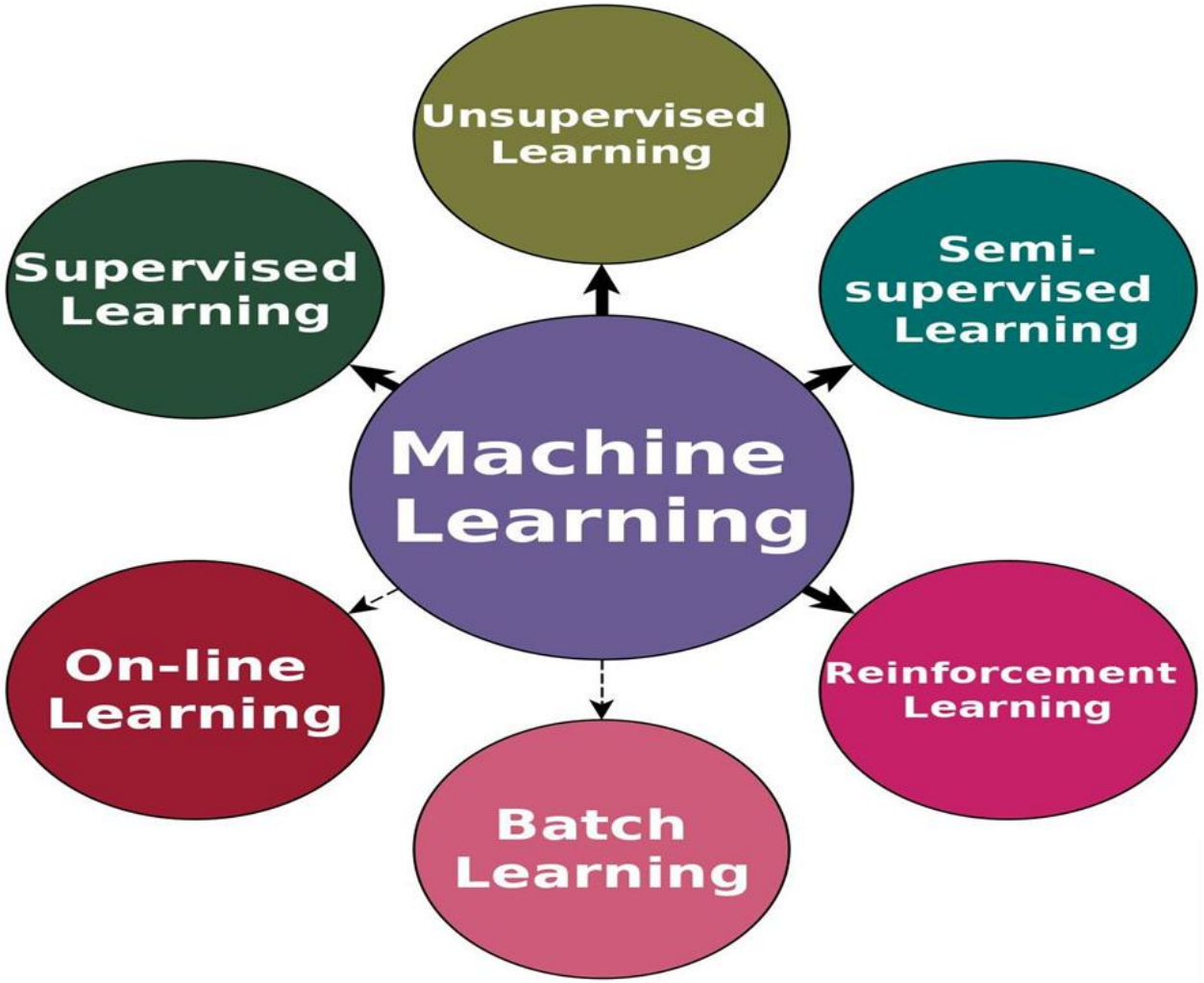
There are so many different types of Machine Learning systems that it is useful to classify them in broad categories based on:

- **Whether or not they are trained with human supervision (supervised, unsupervised,semisupervised,and Reinforcement Learning).**

- **Whether or not they can learn incrementally on the fly (online versus batch learning).**

- **Whether they work by simply comparing new data points to known data points, or instead detect patterns in the training data and build a predictive model, much like scientists do (instance-based versus model- based learning).**

Types of Machine Learning Systems



The types of machine learning

Types of Machine Learning Systems

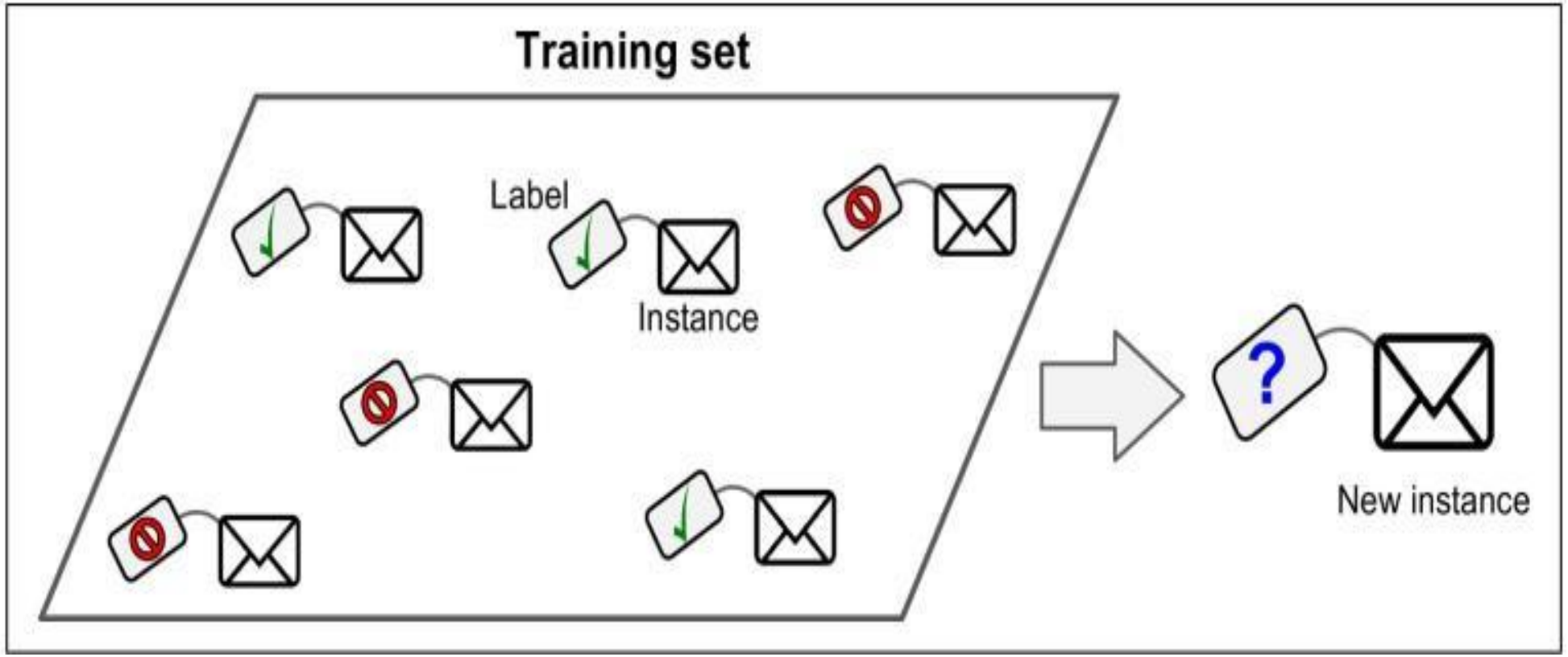
Supervised learning :

In supervised learning, the training data you feed to the algorithm includes the desired solutions, called labels.

$$\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \cdots, (\mathbf{x}^{(n)}, y^{(n)})\}$$

- **When *y* is real, we talk about *regression*.**
- **When *y* is discrete, we talk about *classification*.**

Types of Machine Learning Systems



A labeled training set for supervised learning.

Types of Machine Learning Systems

Here are some of the most important supervised learning algorithms:

- **k-Nearest Neighbors (KNN)**
- **Linear Regression**
- **Logistic Regression**
- **Support Vector Machines (SVMs)**
- **Decision Trees and Random Forests**
- **Neural networks***

Types of Machine Learning Systems

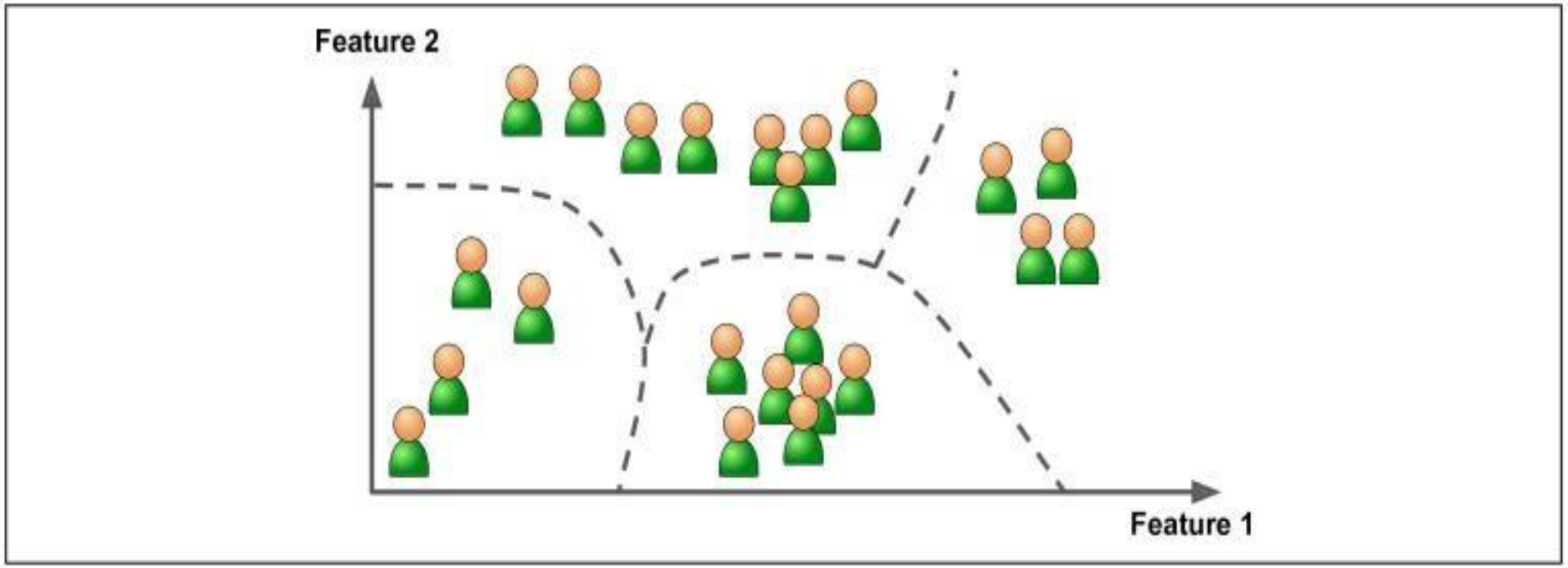
Unsupervised Learning:

In unsupervised learning, as you might guess, the training data is unlabeled. The system tries to learn without a teacher.

$$\mathcal{D} = \{x^{(1)}, x^{(2)}, \cdots, x^{(n)}\}$$

No labels are given to the learning algorithm, leaving it on its own to explore or find structure in the data.

Types of Machine Learning Systems



An unlabeled training set for unsupervised learning.

Types of Machine Learning Systems

Here are some of the most important unsupervised learning algorithms:

- **Clustering**
- **Visualization and dimensionality reduction**

Types of Machine Learning Systems

Semi-Supervised Learning :

Some algorithms can deal with partially labeled training data, usually a lot of unlabeled data and a little bit of labeled data. This is called semi-supervised learning.

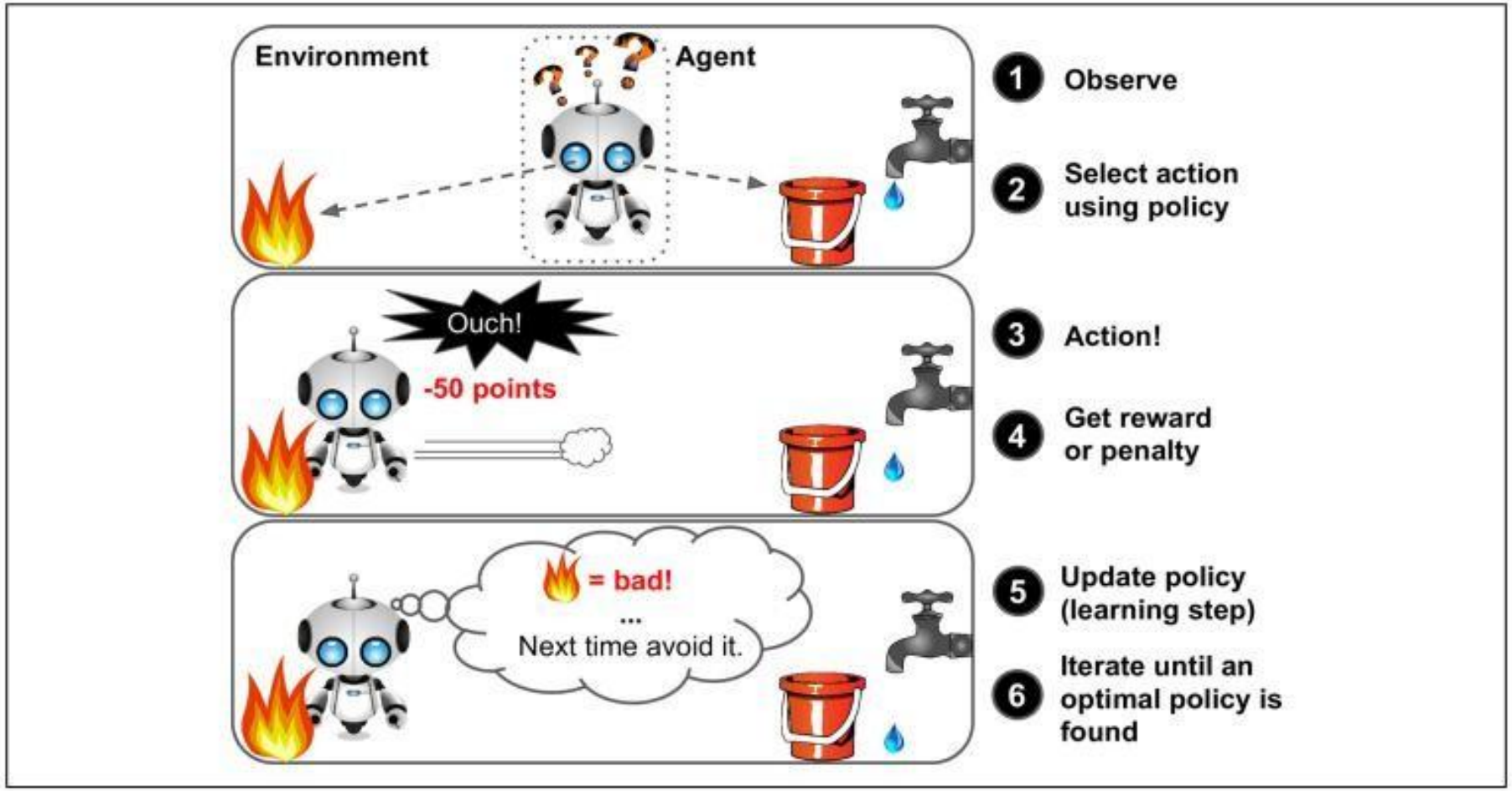
Most semi- supervised learning algorithms are combinations of unsupervised and supervised algorithms.

Types of Machine Learning Systems

Reinforcement Learning :

- **The learning system called an agent in this context.**
- **Can observe the environment, select and perform actions, and get rewards in return (or penalties in the form of negative rewards).**
- **It must then learn by itself what is the best strategy, called a policy, to get the most reward over time.**
- **A policy defines what action the agent should choose when it is in a given situation.**

Types of Machine Learning Systems



Reinforcement Learning

Types of Machine Learning Systems

Batch learning:

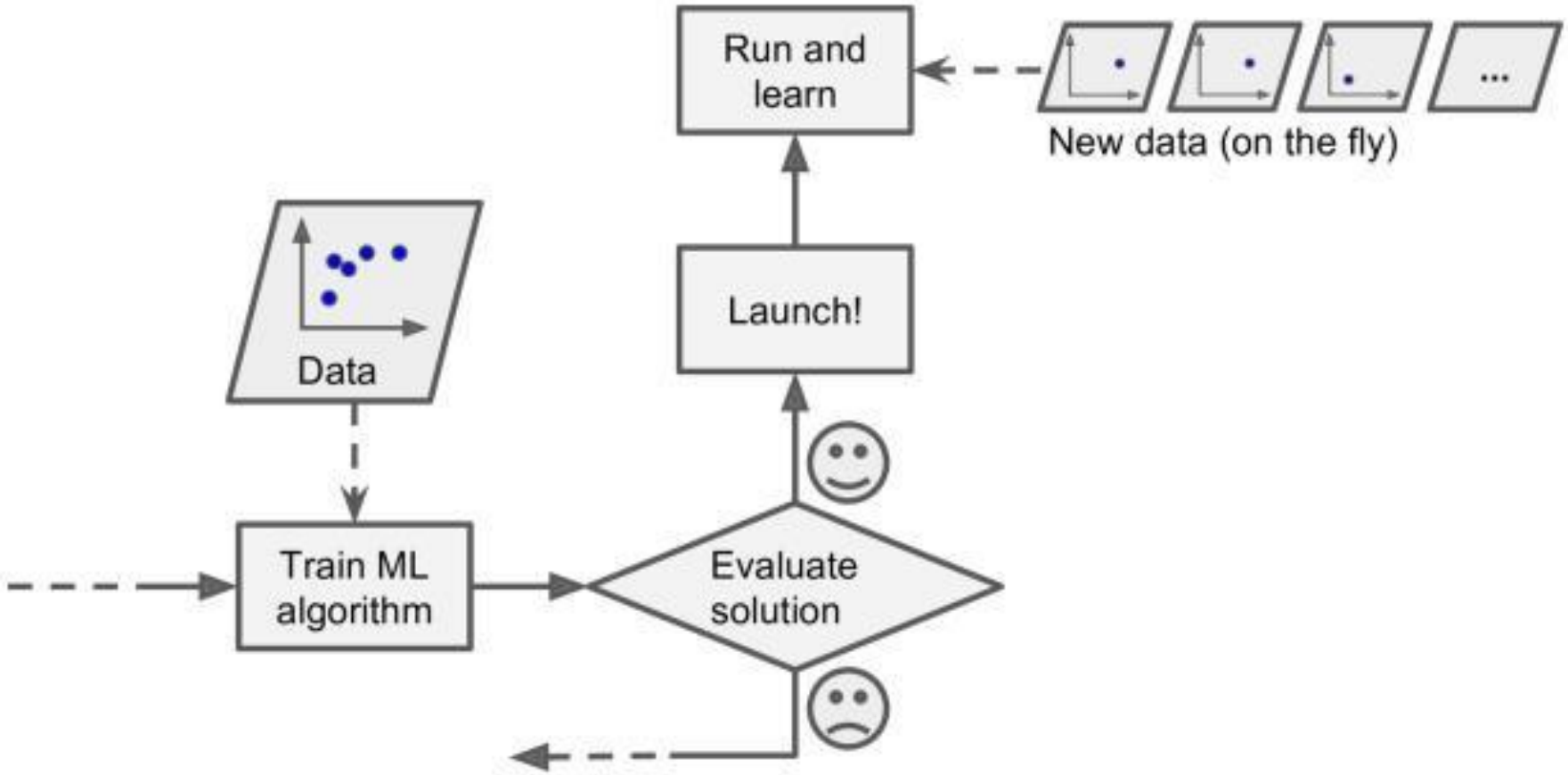
In batch learning, the system is incapable of learning incrementally: **it must be trained using all the available data. This will generally take a lot of time and computing resources, so it is typically done offline. First, the system is trained, and then it is launched into production and runs without learning anymore; it just applies what it has learned. This is called offline learning.**

Types of Machine Learning Systems

On-line learning:

In online learning, you train the system incrementally by feeding it data instances sequentially, either individually or by small groups called mini batches. Each learning step is fast and cheap, so the system can learn about new data on the fly, as it arrives.

Types of Machine Learning Systems



Online learning

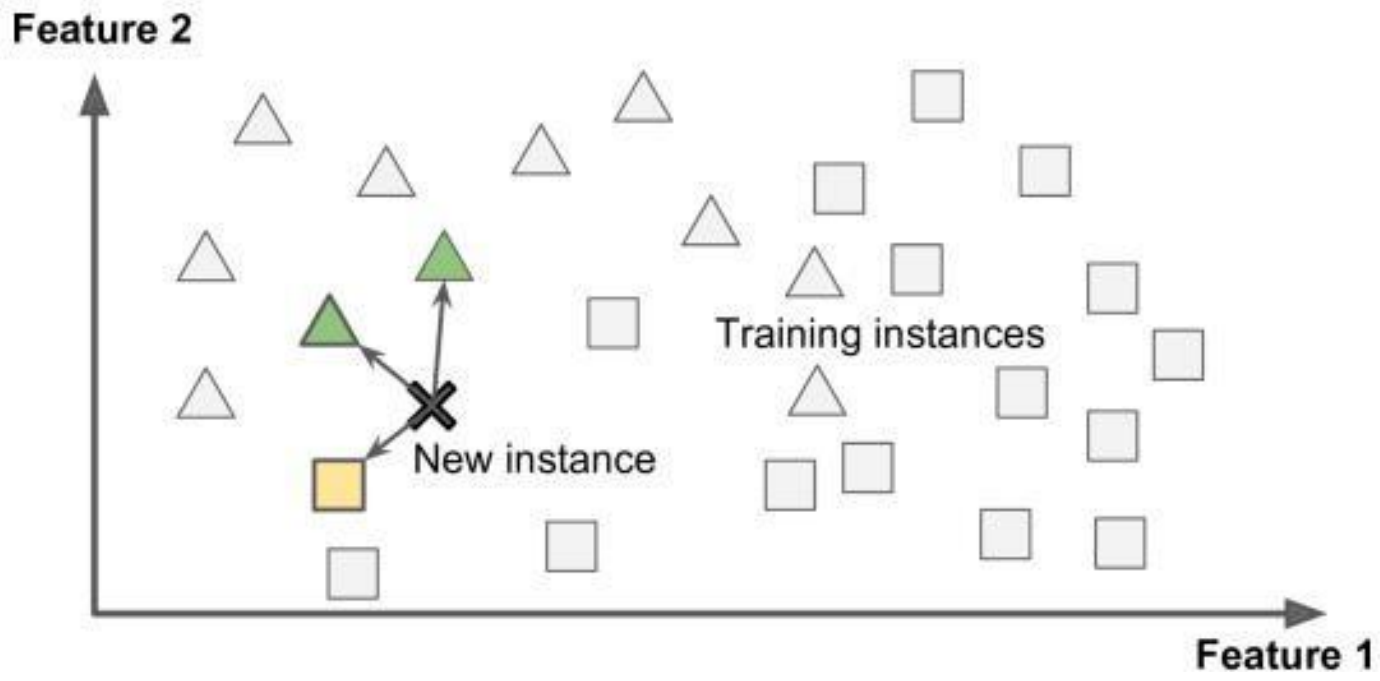
Instance-Based VS Model-Based Learning

One more way to categorize Machine Learning systems is by how they generalize. Most Machine Learning tasks are about making predictions. This means that given a number of training examples, the system needs to be able to generalize to examples it has never seen before.

Having a good performance measure on the training data is good, but insufficient; the true goal is to perform well on new instances. There are two main approaches to generalization: instance-based learning and model-based learning.

Instance-Based VS Model-Based Learning

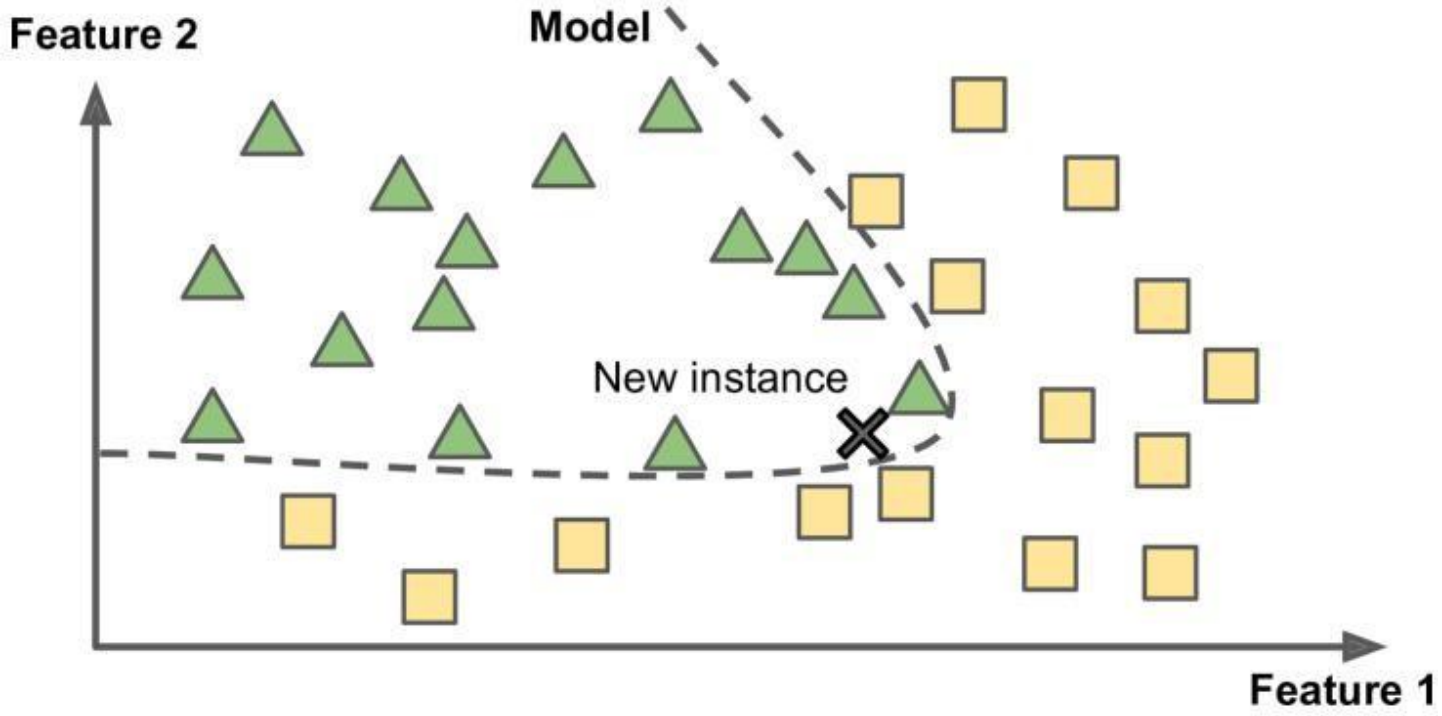
Instance-based learning:
The system learns the examples by heart, then generalizes to new cases using a similarity measure.



Instance-Based VS Model-Based Learning

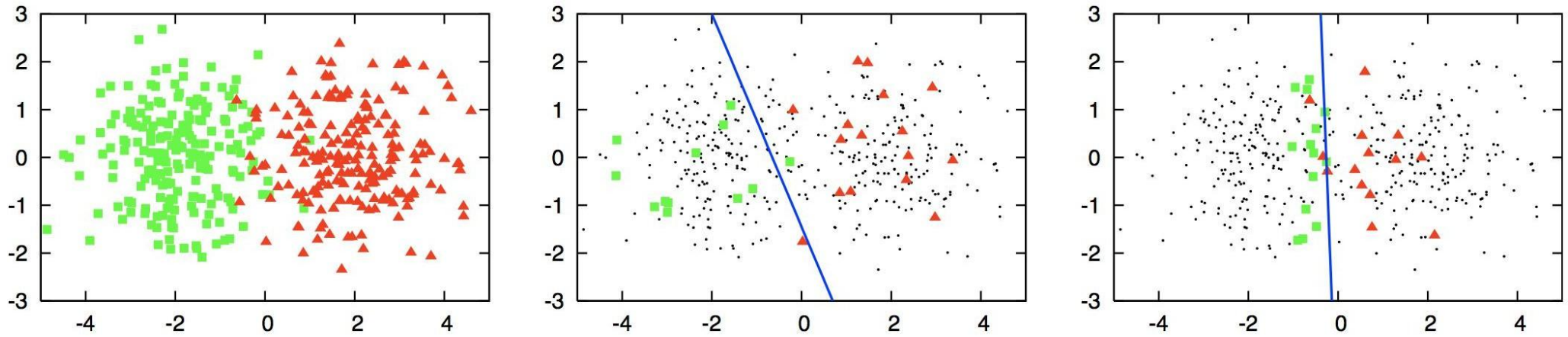
Model-based learning:

Build a model of these examples, then use that model to make predictions.



Active Learning vs Transfer Learning

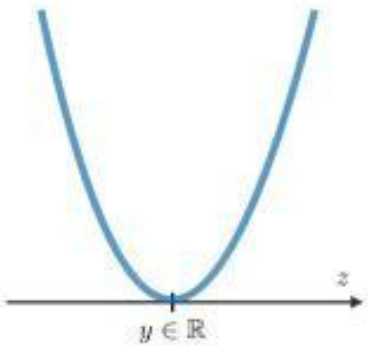
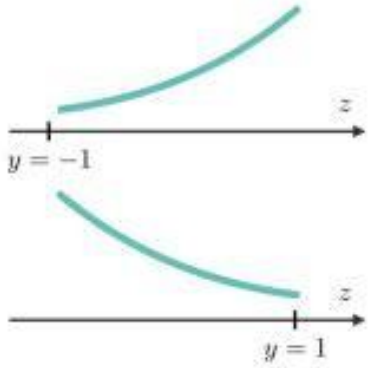
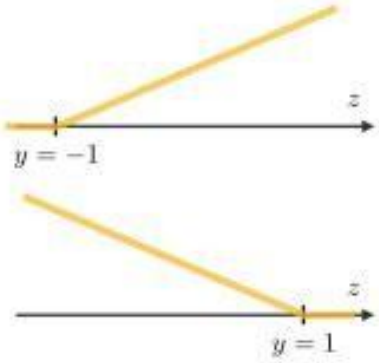
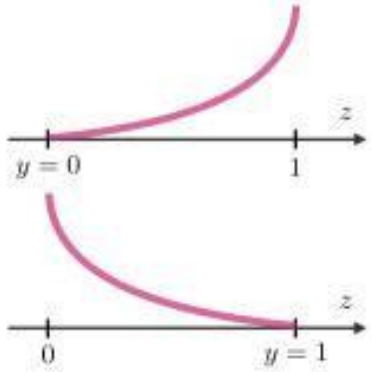
- **Active Learning is a case of semi-supervised machine learning.**
- **For classification, you would want to sample a small subset of examples and find those labels and use these labeled examples as your training data for a classifier.**



Active Learning vs Transfer Learning

Transfer learning (TL) is a research problem in ML that focuses on storing knowledge gained while solving one problem and applying it to a different but related problem. For example, knowledge gained while learning to recognize Cats could apply when trying to recognize Tigers.

Loss Function

Least squared	Logistic	Hinge	Cross-entropy
$\frac{1}{2}(y-z)^2$	$\log(1+\exp(-yz))$	$\max(0,1-yz)$	$-\left[y\log(z)+(1-y)\log(1-z)\right]$
			
Linear regression	Logistic regression	SVM	Neural Network

The loss function computes the error for a single training example, while the cost function is the average of the loss functions of the entire training set.

Machine Learning Vocabulary 3

- **Hyperparameters :** are configuration variables that are external to the model and whose values cannot be estimated from data. That is to say, they can not be learned directly from the data in standard model training. They are almost always specified by the machine learning engineer prior to training.

- **Regression:** this is the problem of predicting a real value for each item. Examples of regression include prediction of stock values or that of variations of economic variables.

- **Classification:** this is the problem of assigning a category to each item.

- **Clustering:** this is the problem of partitioning a set of items into homogeneous subsets.

In Summary

1) **You studied the data.**

2) **You selected a model.**

3) **You trained it on the training data.**

4) **Finally, you applied the model to make predictions on new cases.**

Main Challenges of Machine Learning

In short, since your main task is to select a learning algorithm and train it on some data, the two things that can go wrong are "bad data" and "bad algorithm".

Main Challenges of Machine Learning

2- Algorithm

1) Overfitting the Training Data:

Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data. This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model. The problem is that these concepts do not apply to new data and negatively impact the models ability to generalize.

The model performs well on the training data, but it does not generalize well.

Main Challenges of Machine Learning

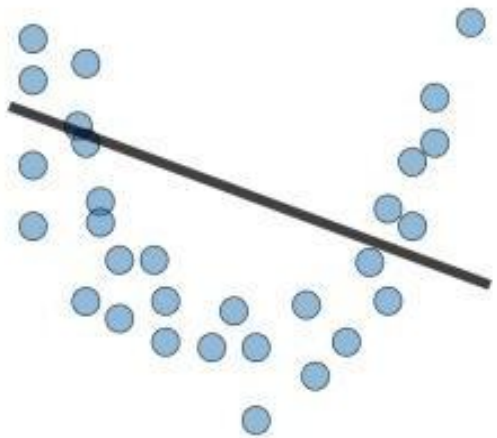
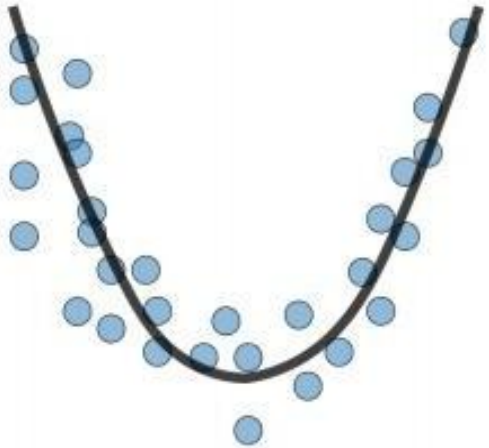
2- Algorithm

2) Underfitting the Training Data:

Underfitting is the opposite of overfitting: it occurs when your model is too simple to learn the underlying structure of the data.

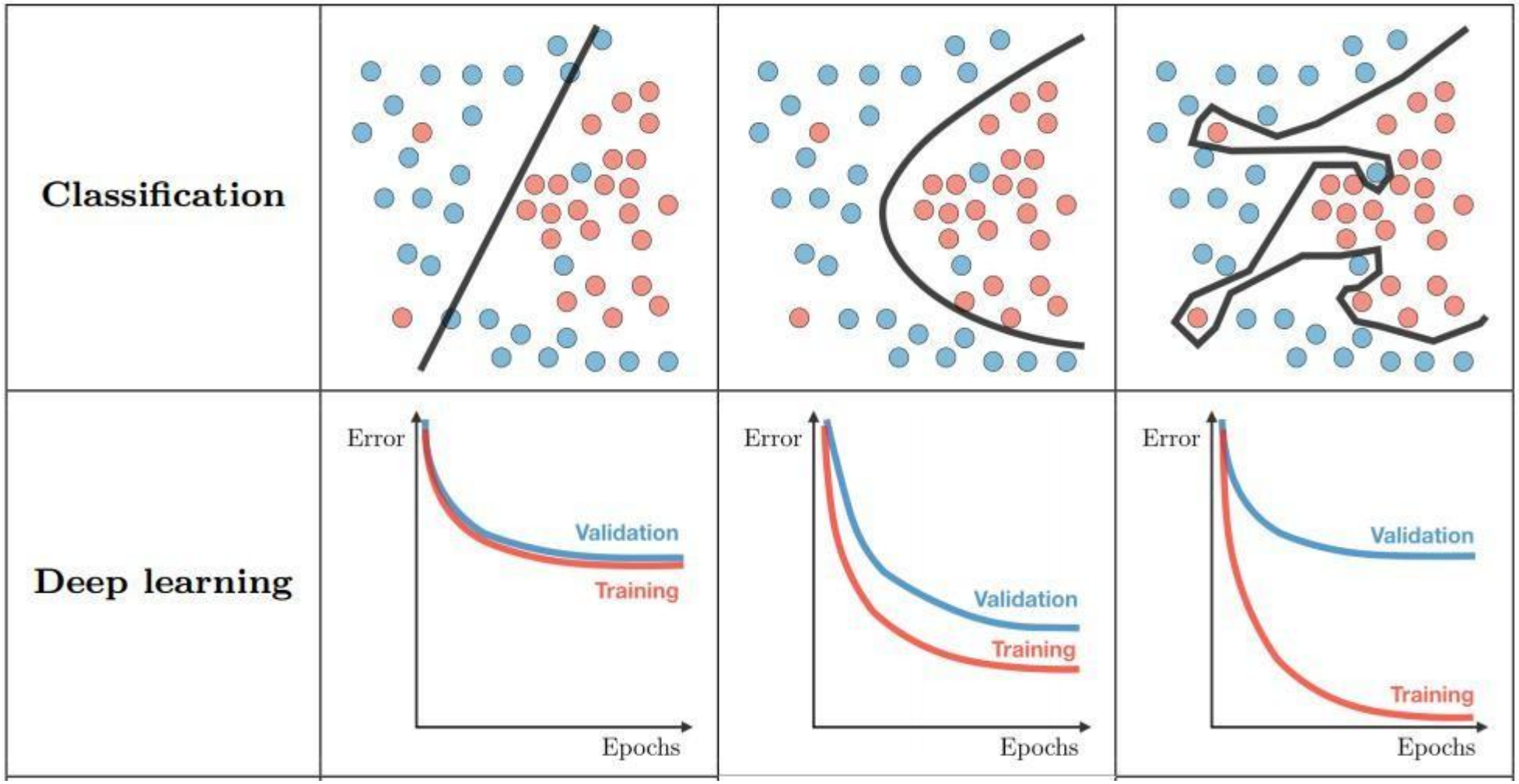
Main Challenges of Machine Learning

2- Algorithm

	Underfitting	Just right	Overfitting
Symptoms	- High training error - Training error close to test error - High bias	- Training error slightly lower than test error	- Low training error - Training error much lower than test error - High variance
Regression			

Main Challenges of Machine Learning

2- Algorithm



How to Avoid Underfitting and Overfitting

Underfitting :

- **Complexify model**
- **Add more features**
- **Train longer**

Overfitting :

- **validation**
- **Perform regularization**
- **Get more data**
- **Remove/Add some features**

Common Classification Model Evaluation Metrics : Confusion Matrix

The confusion matrix is used to describe the performance of a classification model on a set of test data for which true values are known.

		Predicted class	
		+	−
Actual class	+	**TP** True Positives	**FN** False Negatives Type II error
	−	**FP** False Positives Type I error	**TN** True Negatives

Common Classification Model Evaluation metrics :

Main Metrics

Metric	Formula	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall performance of model
Precision	$\frac{TP}{TP + FP}$	How accurate the positive predictions are
Recall Sensitivity	$\frac{TP}{TP + FN}$	Coverage of actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Coverage of actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	Hybrid metric useful for unbalanced classes

Common Classification Model Evaluation metrics :

Main Metrics

Metric	Formula	Equivalent
True Positive Rate TPR	$\frac{TP}{TP + FN}$	Recall, sensitivity
False Positive Rate FPR	$\frac{FP}{TN + FP}$	1-specificity

Common Regression Model Evaluation metrics : Mean Absolute Error

$$MAE = \boxed{\frac{1}{n}} \sum \left| \boxed{y} - \boxed{\hat{y}} \right|$$

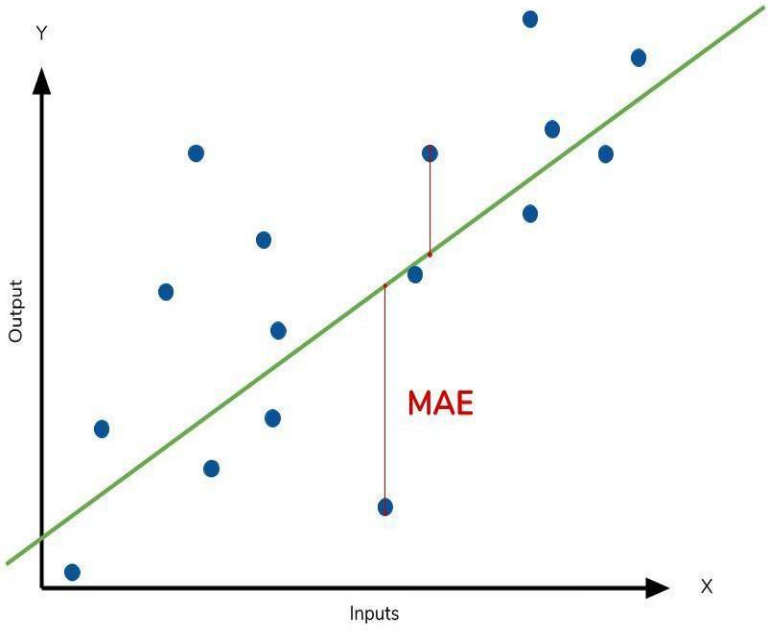
Divide by the total number of data points

Actual output value

Predicted output value

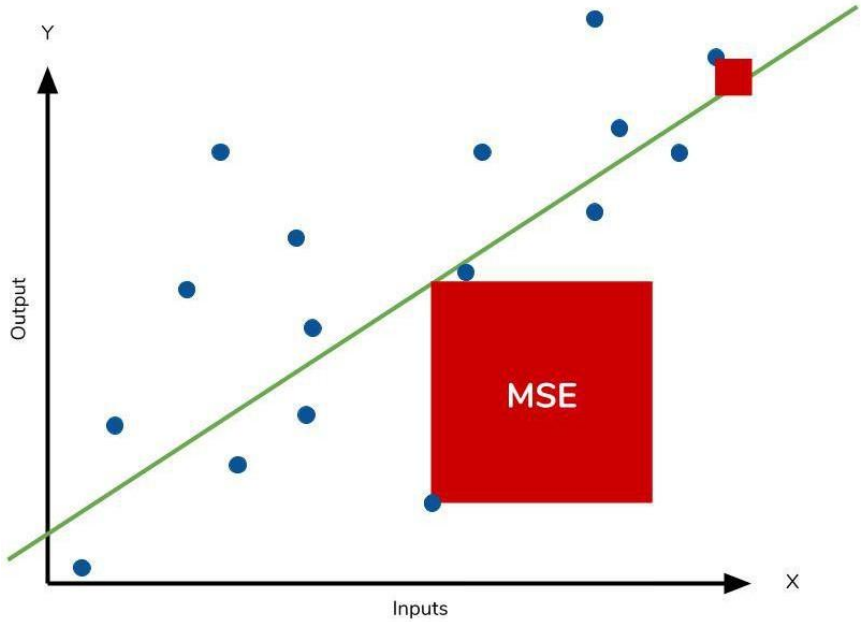
Sum of

The absolute value of the residual



Common Regression Model Evaluation metrics : Mean Square Error

$$MSE = \frac{1}{n} \Sigma \underbrace{\left(y - \hat{y} \right)}_{\text{The square of the difference between actual and predicted}}^2$$

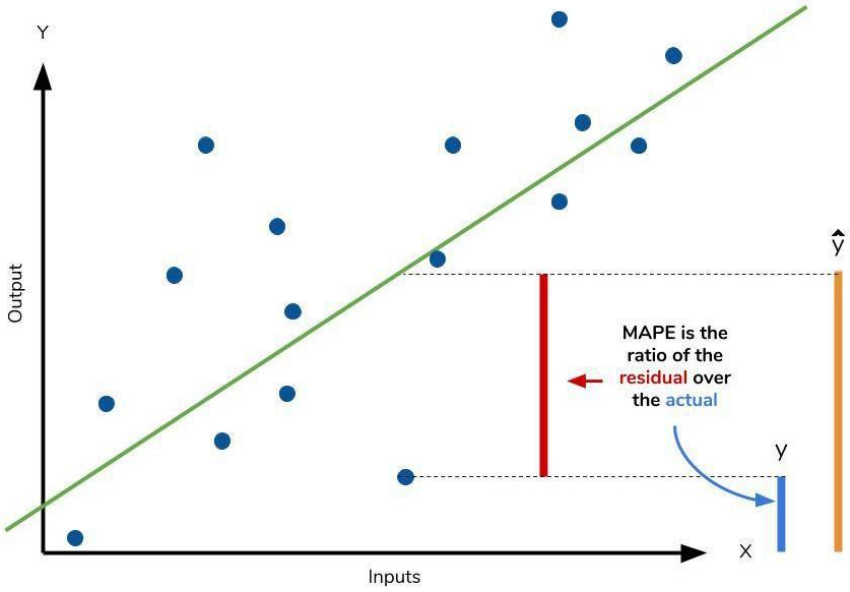


Common Regression Model Evaluation metrics : Mean Absolute Percentage Error

Multiplying by 100% converts to percentage

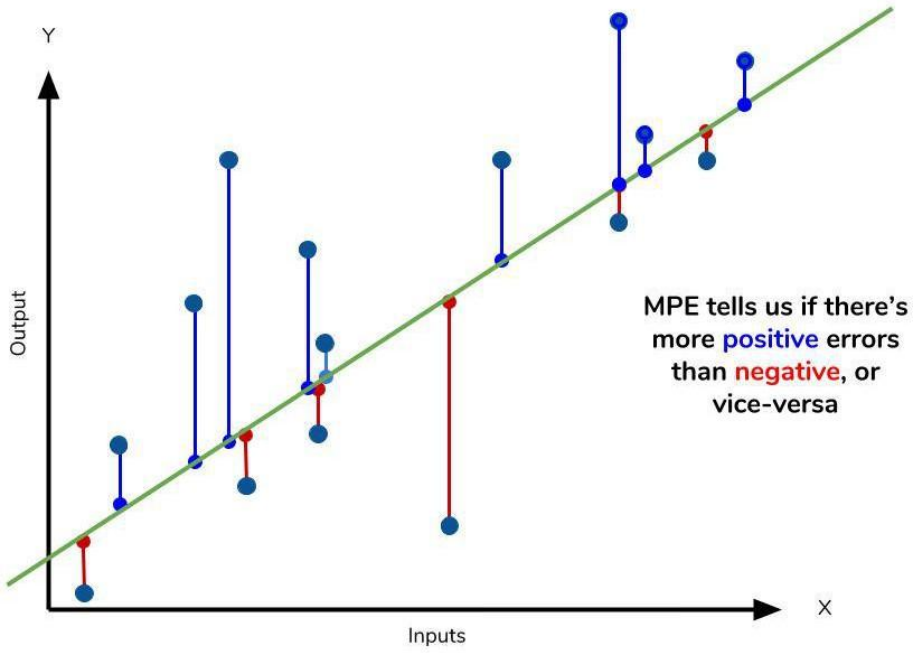
$$MAPE = \frac{\boxed{100\%}}{n} \sum \left| \frac{\overbrace{y - \hat{y}}^{\text{The residual}}}{\boxed{y}} \right|$$

Each residual is scaled against the actual value



Common Regression Model Evaluation metrics : Mean Percentage Error

$$MPE = \frac{100\%}{n} \Sigma \left(\frac{y - \hat{y}}{y} \right)$$



Testing and Validating

It is common to use 80% of the data for training and hold out 20% for testing.

If the training error is low (i.e., your model makes few mistakes on the training set) but the generalization (testing) error is high, it means that your model is overfitting the training data.

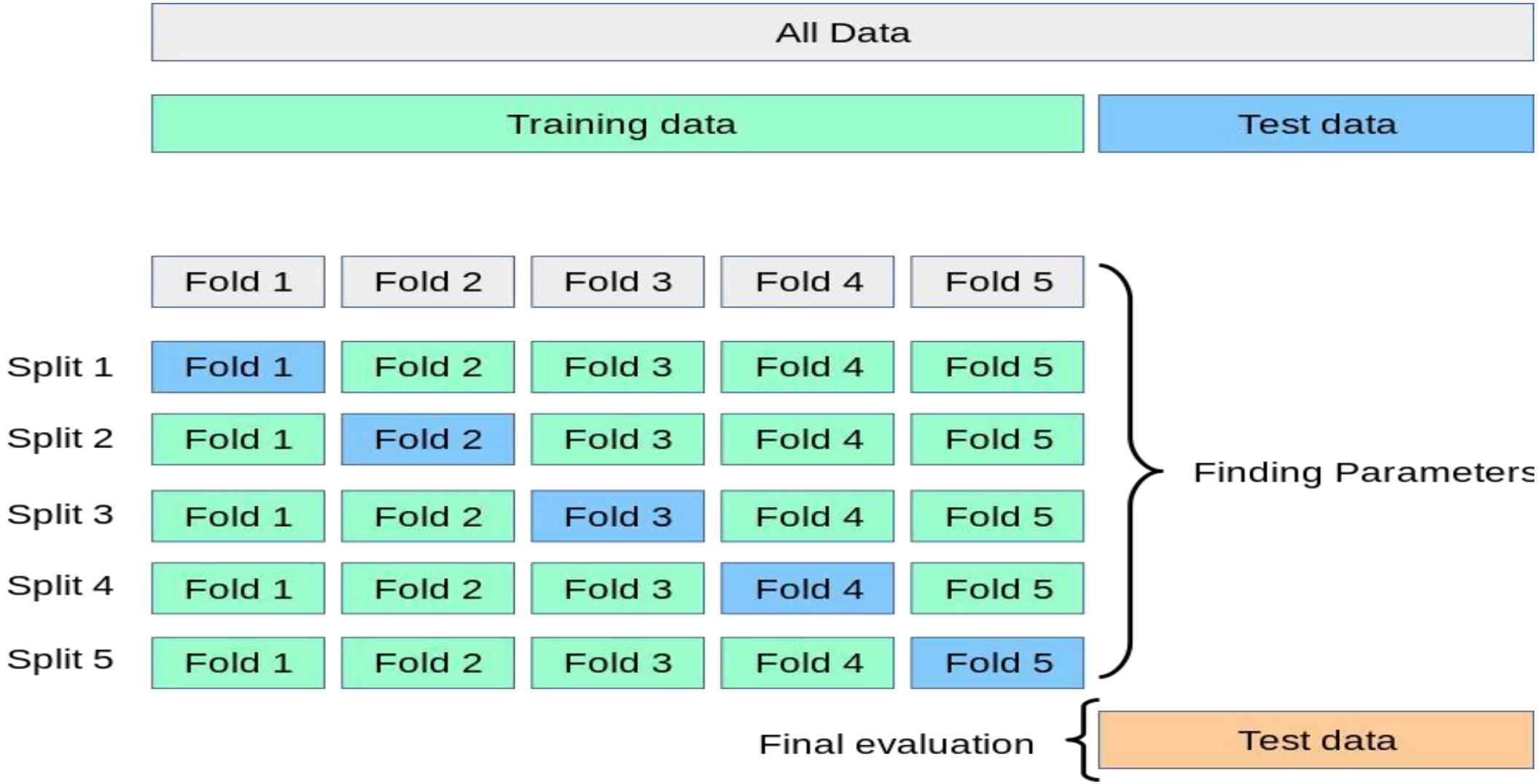
A common solution to this problem is to have a second holdout set called the validation set.
You train multiple models with various hyperparameters using the training set, you select the model and hyperparameters that perform best on the validation set, and when you're happy with your model you run a single final test against the test set to get an estimate of the generalization error.

Testing and Validating : Cross-Validation

Cross-Validation (CV) : the training set is split into complementary subsets, and each model is trained against a different combination of these subsets and validated against the remaining parts.

Once the model type and hyperparameters have been selected, a final model is trained using these hyperparameters on the full training set, and the generalized error is measured on the test set.

Testing and Validating : Cross-Validation

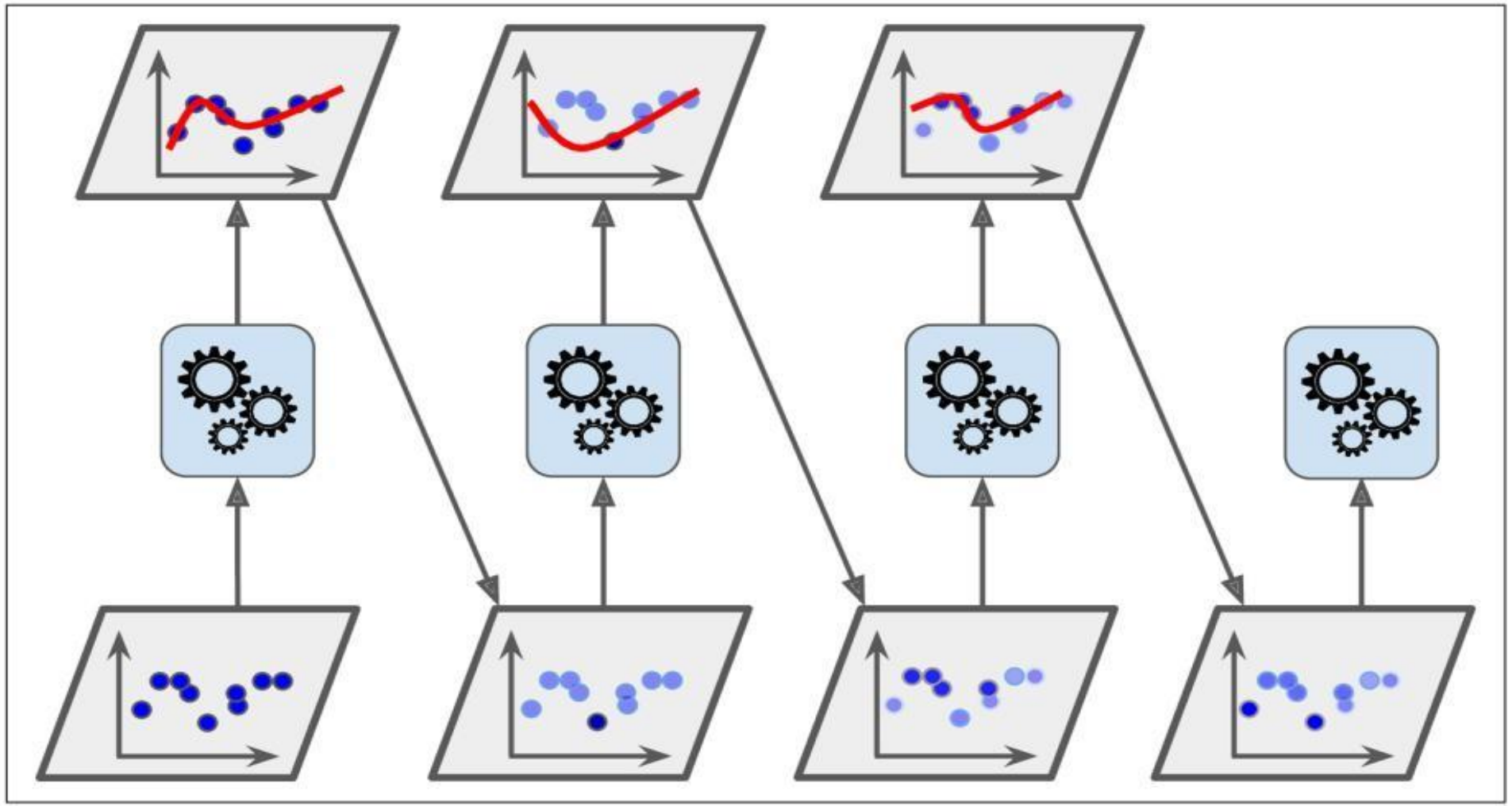


Boosting

Boosting refers to any Ensemble method that can combine several weak learners into a strong learner.

The general idea of most boosting methods is to train predictors sequentially, each trying to correct i ts predecessor. There are many boosting methods available, but by far the most popular are AdaBoost (Adaptive Boosting) and Gradient Boosting.

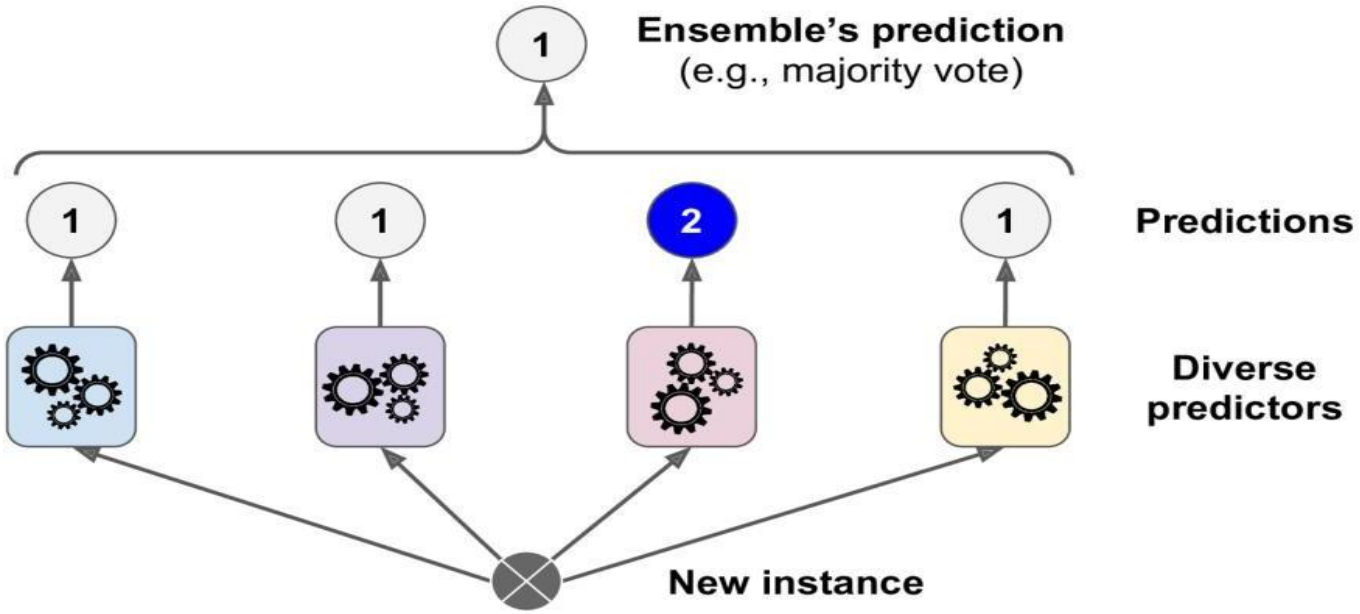
Boosting



AdaBoost sequential training with instance weight updates

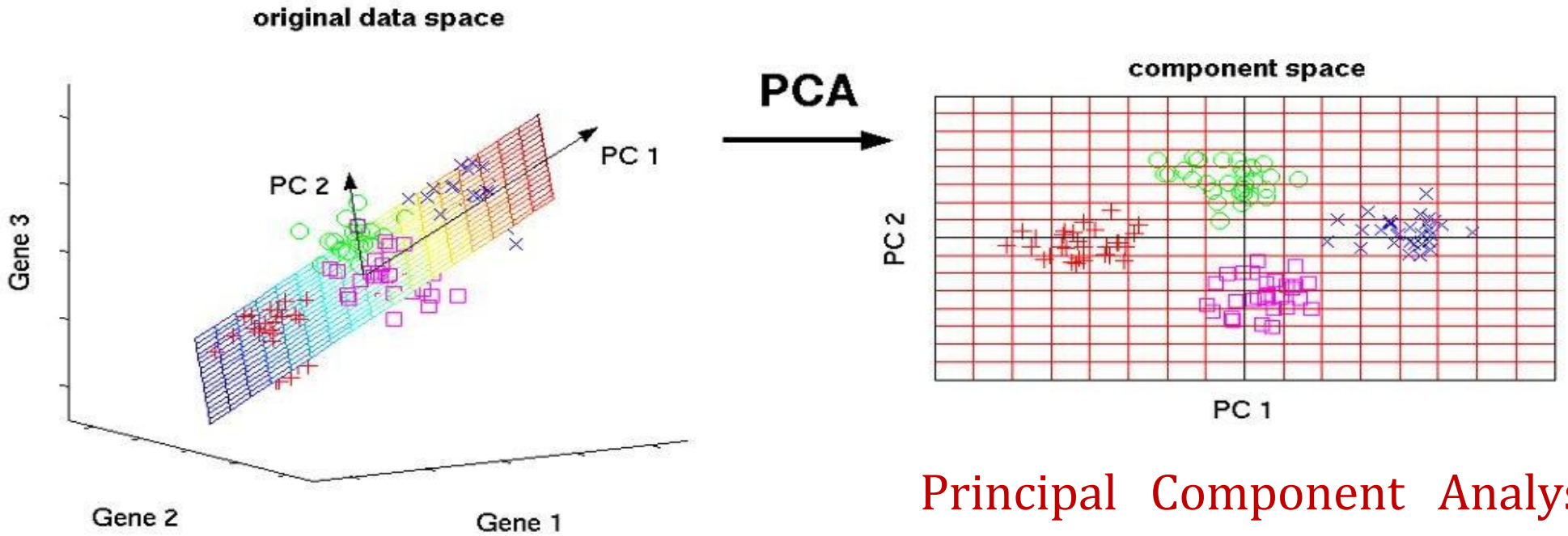
Voting Classifiers

The Voting Classifier: **is a meta-classifier for combining similar or conceptually different machine learning classifiers for classification via majority or plurality voting. (For simplicity, we will refer to both majority and plurality voting as majority voting.)**



Dimensionality Reduction

Many Machine Learning problems involve thousands or even millions of features for each training instance. Not only does this make training extremely slow, but it can also make it much harder to find a good solution. This problem is often referred to as the curse of dimensionality.



Principal Component Analysis

Hyperparameter Tuning

Hyperparameter Tuning : works by running multiple trials in a single training job. Each trial is a complete execution of your training application with values for your chosen hyperparameters, set within limits you specify. The AI Platform training service keeps track of the results of each trial and makes adjustments for subsequent trials. When the job is finished, you can get a summary of all the trials along with the most effective configuration of values according to the criteria you specify.

Steps to Build a Machine Learning System

1. **Data collection.**
2. **Improving data quality (data preprocessing: drop duplicate rows, handle missing values and outliers).**
3. **Feature engineering (feature extraction and selection, dimensionality reduction).**
4. **Splitting data into training (and evaluation) and testing sets.**
5. **Algorithm selection (Regression, Classification, Clustering ...).**
6. **Training.**
7. **Evaluation + Hyperparameter tuning.**
8. **Testing.**
9. **Deployment**

Thank you for your attention